An assigned equality with a fixed anchor

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be an ambient string action, put $H = G_{\geq k}$ and $K = G_{>k}$, and let V be an actual H-orbit. Eligibility refers throughout to the full ambient omissions G_j.

Theorem 1. *Suppose $x \in V$ is fixed by K. Let $F \subseteq \{k, \dots, r-1\}$ contain k, and suppose every colour in F has two ambient-eligible edges in V. If $|V| = 2|F|$, then F is one initial interval $\{k, \dots, a-1\}$ for some a.*

Proof. Ambient eligibility implies eligibility for the full local omission of $H|_V$, by subgroup inclusion. The complete selected-colour equality theorem therefore makes G_F transitive on V.

Split F into its maximal consecutive intervals. Their generated groups commute pairwise, since labels in different intervals differ by at least two. Every component except the one containing k lies in K and fixes x. In a transitive product of commuting permutation groups, a factor fixing x fixes every point: write an element as ab, with a in that factor and b in the product of the others. Then $x^{ab} = x^b$, and the whole factor fixes x^b by commutation with b. Thus every later component fixes all of V. But every one of its colours has two actual edges in V, a contradiction. Only the initial component remains. □

Corollary 2. *The entire three-orbit budget of the acquired theorem* A four-point internal orbit gives a nonterminal separator *is impossible.*

Proof. Retain every literal hypothesis of that theorem, including the natural alternating tail, full IP, $n = 2r+1$, seven $H = G_{\geq 5}$-fixed points, nontrivial orbits E, W, V of sizes $2, 4, |V|$, the specified action of δ, and intransitivity of $K = G_{>5}$ on W. Its assignments are

$$F_E = \varnothing, \quad F_W = \{j\}, \quad F_V = \{5, \dots, r-1\} \setminus \{j\}, \qquad j \geq 6.$$

The same actual head argument supplies $x \in V$ fixed by K. Moreover

$$|V| = n - 7 - 2 - 4 = 2r - 12 = 2|F_V|.$$

The theorem just proved makes F_V an initial interval, so its only missing tail label j must be $r-1$. The complete acquired four-point-row theorem, with exactly these hypotheses, gives $j < r-1$. This contradiction excludes the budget in full. □

Attribution and remaining scope. The selected-colour equality and complete four-point-row proof are acquired inputs, with every predecessor retained. The new observation is the action of commuting marked components on a fixed anchor. It uses no projected IP, new computation or rank induction. In the remaining nonfirst internal deficit-two budget it forces the first assigned set to be an initial interval and the other assigned set to be terminal; it does not exclude that budget. Split-orbit pairs, global attachment and full L044 recovery remain. No formal, original, question or numerical credit is claimed.